in Fig. 27G(*a*). The compensator C is oriented at about 45° with N_1 and its image focused on a screen MM by means of a lens L_2. Owing to the effectively variable thickness along the compensator, the light on the screen (with N_2 removed) will be polarized as shown in Fig. 27G(*b*) (see also Sec. 27.1 and Fig. 12K). If a second nicol is inserted at N_2 and oriented perpendicular to one set of plane-polarization regions, e.g., those marked 1, 2, and 3 in the figure, no light will be transmitted at these points. The screen will thus be crossed by a set of equidistant parallel dark bands. With white light the bands are colored and look like Young's double-slit fringes but with the center one black. Best results are obtained with monochromatic light. Piles of glass plates mounted in tubes or Polaroid films may of course be used in place of nicols N_1 and N_2.

27.5 ANALYSIS OF POLARIZED LIGHT

If one has a beam of light that is completely polarized, linearly, elliptically, or circularly, it will appear to the eye no different from ordinary unpolarized light. By using simple auxiliary apparatus, however, its character and vibration form can easily be determined. For this purpose an analyzer, in the form of a nicol or Polaroid, is used in conjunction with either a quarter-wave plate or some form of compensator. For many purposes a quarter-wave plate is adequate, and the compensator need be used only where precise measurements of elliptical polarization are required.

To illustrate the use of the quarter-wave plate, suppose, for example, that it is placed in a beam of circularly polarized light. Regardless of the orientation of the optic axis, the circular vibration is equivalent to two linear and mutually perpendicular ones along the slow and fast axes, 90° out of phase with each other. Upon emerging from the plate these two are in phase and recombine to give plane-polarized light vibrating at 45° with the axes of the plate. The plane of the emergent light depends on the direction of rotation of the incident circularly polarized light. In either of the possible cases it can be completely extinguished by the analyzer. If the light to be studied is elliptically polarized, it will be converted into plane-polarized light only when the fast axis of the quarter-wave plate coincides with either the major or the minor axis of the ellipse. The ratio of these axes can then be found as the tangent of the angle that the plane of transmission of the analyzer makes with the fast axis when extinction has been achieved.

The same information may be found with greater accuracy by means of a Babinet compensator, which has the additional advantage of being usable for any wavelength. We have seen that when the incident light is plane-polarized at 45° to the principal section of one wedge, a dark band occurs at the center. If for some other kind of light the dark band is displaced from this position, a phase difference must exist between the two rectangular components of this light, and this means that it is elliptical to a greater or less degree. Since a phase difference of 2π corresponds to one whole fringe, the actual difference can be found from the fractional fringe displacement. The measurement is done by screwing one wedge over the other until the dark fringe returns to the center, so that the phase difference has been *compensated*.

For details of the use of the compensator, the reader should refer to more advanced texts.*

If the light is not completely polarized but contains some admixture of unpolarized light, it is still possible to completely determine its character by using a quarter-wave plate and an analyzer in the systematic way outlined in Table 27A. The light is first studied with the analyzer alone. If there is no change in intensity on rotating it, the procedure outlined in part A of the table is followed. If there is some change of intensity, part B is followed. The seven kinds of light which can be identified in this way represent all the possible conditions of polarization. Other more complicated mixtures can be shown to be equivalent to one or another of these.

To specify quantitatively the state of polarization of a beam of light, it is found that just four numbers are required. These *Stokes parameters* can be determined by making four suitably chosen measurements. One of these involves the total intensity, and another requires some phase-shifting device like a quarter-wave plate in conjunction with an analyzer. The remaining two can be made with the analyzer alone.†

27.6 INTERFERENCE WITH WHITE LIGHT

Referring to Eq. (27b), it is observed that the phase difference between the E and O rays depends upon the wavelength as well as on the thickness of the plate. As for the difference between the principal indices of refraction $(n_O - n_E)$, the values given in Table 26A show that there is little change throughout the visible region. As the thickness of a crystal plate increases, the phase difference δ between the O and E rays for violet light, $\lambda 4000$, increases nearly twice as fast as the phase difference for red light, $\lambda 6500$, since λ occurs in the denominator of the expression for δ. This fact gives rise to the rich colors frequently observed in thin plates of mica, quartz, calcite, etc., cut parallel to the axis and placed between crossed nicols. The reason for the color is that some one or more parts of the continuous visible spectrum are stopped by the second nicol prism.

Suppose that a thin sheet of mica which introduces a phase change for yellow light of 2π rad, that is, a full-wave plate, is introduced between crossed nicols and oriented at an angle $\theta = 45°$. The orange and red wavelengths will undergo a phase change less than 2π and the green, blue, and violet more than 2π. Components of all colors but yellow light will therefore get through the second nicol. With yellow absent, the resultant color will be the mixture of red, orange, green, blue, and violet, giving a purple hue.

If, in the above experiment with a mica sheet, the analyzing nicol is replaced with a thick natural calcite crystal, one obtains the ordinary vibrations O' and O'' as well as the extraordinary ones (Fig. 27H) but in a different position. This O beam is also colored and is complementary to the E beam containing the components

* M. Born, "Optik," p. 244, J. Springer, Berlin, 1933.

† A summary of the uses of the Stokes parameters, including their application to photons and elementary particles, is given by W. H. McMaster, *Am. J. Phys.*, **22**: 351 (1954).

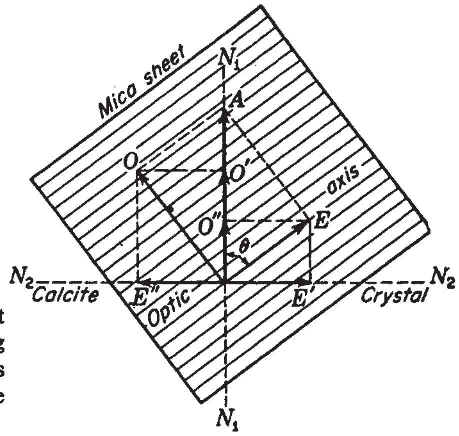

FIGURE 27H
Components of plane-polarized light transmitted by a thin doubly refracting plate and an analyzing crystal. The lines N_1 and N_2 indicate the directions of the O and E vibrations in the calcite.

Table 27A ANALYSIS OF POLARIZED LIGHT

<table>
<tr><td colspan="3">A. No intensity variation with analyzer alone</td></tr>
<tr><td>I. If with $\lambda/4$ plate in front of analyzer</td><td colspan="2">II. If with $\lambda/4$ plate in front of analyzer one finds a maximum, then</td></tr>
<tr><td>1. One has no intensity variation,
one has
natural unpolarized light</td><td>2. If one position of analyzer gives zero intensity,
one has
circularly polarized light</td><td>3. If no position of analyzer gives zero intensity,
one has
mixture of circularly polarized light and unpolarized light</td></tr>
</table>

<table>
<tr><td colspan="4">B. Intensity variation with analyzer alone</td></tr>
<tr><td>I. If one position of analyzer gives</td><td colspan="3">II. If no position of analyzer gives zero intensity</td></tr>
<tr><td rowspan="3">1. Zero intensity,
one has
plane-polarized light</td><td colspan="3">2. Insert a $\lambda/4$ plate in front of analyzer with optic axis parallel to position of maximum intensity</td></tr>
<tr><td rowspan="2">(a) If get zero intensity with analyzer,
one has
elliptically polarized light</td><td colspan="2">(b) If get no zero intensity,</td></tr>
<tr><td>(1) But the same analyzer setting as before gives the maximum intensity,
one has
mixture of plane-polarized light and unpolarized light</td><td>(2) But some other analyzer setting than before gives a maximum intensity,
one has
mixture of elliptically polarized light and plane-polarized light</td></tr>
</table>